Setting up the YOLO Environment

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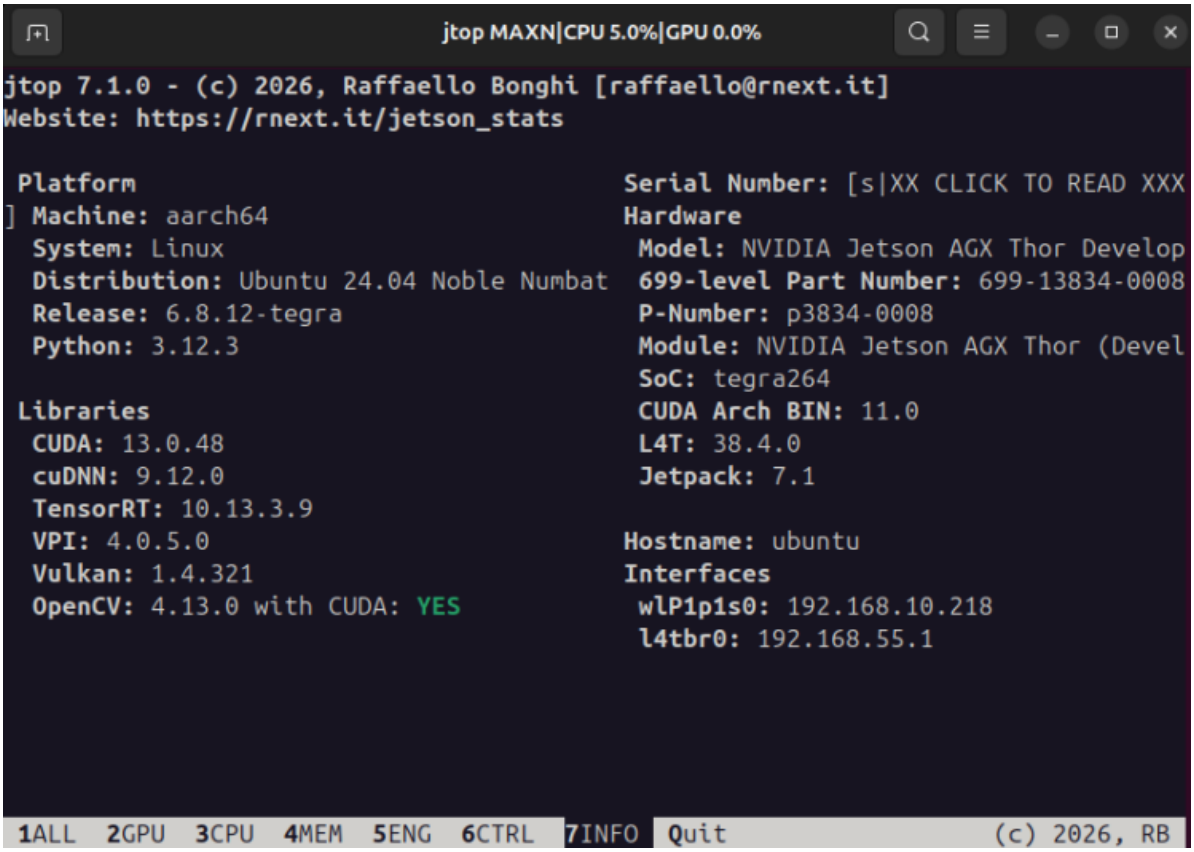
References

1. System Information



2. Downloading Files

This example uses ultralytics-v8.4.21.

```
cd ~/
git clone https://github.com/ultralytics/ultralytics.git -b v8.4.21 --depth=1
```

```
jetson@ubuntu:~$ git clone https://github.com/ultralytics/ultralytics.git -b v8.
4.21 --depth=1
Cloning into 'ultralytics'...
remote: Enumerating objects: 979, done.
remote: Counting objects: 100% (979/979), done.
remote: Compressing objects: 100% (827/827), done.
remote: Total 979 (delta 209), reused 290 (delta 137), pack-reused 0 (from 0)
Receiving objects: 100% (979/979), 2.44 MiB | 59.00 KiB/s, done.
Resolving deltas: 100% (209/209), done.
Note: switching to '83471a5041fe5d7fdba5c12aff65f4f84f27c081'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.
```

If you cannot clone the repository due to network issues, you can download the compressed file from the following website and then extract it.

```
https://github.com/ultralytics/ultralytics/releases/tag/v8.4.21
```

3. Resolving Dependency Libraries

```
sudo pip3 install -U protobuf --ignore-installed --break-system-packages
```

```
jetson@ubuntu:~$ sudo pip3 install -U protobuf --ignore-installed --break-system
-packages
Collecting protobuf
  Using cached protobuf-7.34.0-cp310-abi3-manylinux2014_aarch64.whl.metadata (59
5 bytes)
Using cached protobuf-7.34.0-cp310-abi3-manylinux2014_aarch64.whl (325 kB)
Installing collected packages: protobuf
Successfully installed protobuf-7.34.0
WARNING: Running pip as the 'root' user can result in broken permissions and con
flicting behaviour with the system package manager. It is recommended to use a v
irtual environment instead: https://pip.pypa.io/warnings/venv
jetson@ubuntu:~$
```

```
sudo pip3 install -U packaging --ignore-installed --break-system-packages
```

```
jetson@ubuntu:~$ sudo pip3 install -U packaging --ignore-installed --break-syste
m-packages
Collecting packaging
  Using cached packaging-26.0-py3-none-any.whl.metadata (3.3 kB)
Using cached packaging-26.0-py3-none-any.whl (74 kB)
Installing collected packages: packaging
Successfully installed packaging-26.0
WARNING: Running pip as the 'root' user can result in broken permissions and con
flicting behaviour with the system package manager. It is recommended to use a v
irtual environment instead: https://pip.pypa.io/warnings/venv
jetson@ubuntu:~$
```

```
sudo pip3 install -U attrs --ignore-installed --break-system-packages
```

```
jetson@ubuntu:~$ sudo pip3 install -U attrs --ignore-installed --break-system-pa
ckages
Collecting attrs
  Downloading attrs-25.4.0-py3-none-any.whl.metadata (10 kB)
Downloading attrs-25.4.0-py3-none-any.whl (67 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 67.6/67.6 kB 683.3 kB/s eta 0:00:00
Installing collected packages: attrs
Successfully installed attrs-25.4.0
WARNING: Running pip as the 'root' user can result in broken permissions and con
flicting behaviour with the system package manager. It is recommended to use a v
irtual environment instead: https://pip.pypa.io/warnings/venv
jetson@ubuntu:~$
```

4. Installing Ultralytics

```
cd ~/ultralytics
sudo pip3 install ".[export]" --break-system-packages -i
https://pypi.tuna.tsinghua.edu.cn/simple
```

```
jetson@ubuntu:~$ cd ~/ultralytics
jetson@ubuntu:~/ultralytics$ sudo pip3 install ".[export]" --break-system-packag
es -i https://pypi.tuna.tsinghua.edu.cn/simple
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Processing /home/jetson/ultralytics
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: numpy>=1.23.0 in /usr/lib/python3/dist-packages (
from ultralytics==8.4.21) (1.26.4)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/lib/python3/dist-packag
es (from ultralytics==8.4.21) (3.6.3)
```

5. Configuring GPU Acceleration

Since we have already installed `torch` 2.10.0 and `torchvision` 0.25.0 in the previous tutorial, there is no need to run the `torch` and `torchvision` installation commands here; we only need to install the remaining packages.

> **torch**

> **torchvision**

> **onnxruntime-gpu**

```
sudo pip3 install
https://github.com/ultralytics/assets/releases/download/v0.0.0/onnxruntime_gpu-
1.24.0-cp312-cp312-linux_aarch64.whl --break-system-packages
```

```
jetson@ubuntu:~$ sudo pip3 install https://github.com/ultralytics/assets/release
s/download/v0.0.0/onnxruntime_gpu-1.24.0-cp312-cp312-linux_aarch64.whl --break-s
ystem-packages
Collecting onnxruntime-gpu==1.24.0
```

6. Verify Installation

> Verify Ultralytics

```
python3 -c "import ultralytics; print(ultralytics.__version__)"
```

> Verify Torch

```
python3 -c "import torch; print(torch.__version__);
print(torch.cuda.is_available())"
```

> Verify Torchvision

```
python3 -c "import torchvision; print(torchvision.__version__)"
```

> Verify NumPy

```
python3 -c "import numpy; print(numpy.__version__)"
```

```
jetson@ubuntu:~/ultralytics$ python3 -c "import ultralytics; print(ultralytics._
_version__)"
8.4.21
jetson@ubuntu:~/ultralytics$ python3 -c "import torch; print(torch.__version__);
 print(torch.cuda.is_available())"
2.10.0
True
jetson@ubuntu:~/ultralytics$ python3 -c "import torchvision; print(torchvision._
_version__)"
0.25.0
jetson@ubuntu:~/ultralytics$ python3 -c "import numpy; print(numpy.__version__)"
1.26.4
jetson@ubuntu:~/ultralytics$
```

> Verify onnxruntime-gpu

```
pip3 list | grep onnx
```

```
jetson@ubuntu:~$ pip list | grep onnx
onnx                    1.20.1
onnxruntime-gpu         1.24.0
onnxslim                0.1.87
jetson@ubuntu:~$
```

References

https://docs.ultralytics.com/guides/nvidia-jetson/